

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Gene Expression (GO:0010467)	-0.10052218	264	2.167e-08	1.168e-04	NLRP3:2 DEDD2:35 MRPS18C:71 NCBP:106 RPS13:196 MRPL34:202
mRNA Processing (GO:0006397)	-0.11420035	190	6.250e-08	1.684e-04	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
RNA Splicing, Via Transesterification Re	-0.11903688	160	2.189e-07	9.932e-04	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
mRNA Splicing, Via Spliceosome (GO:00003	-0.10285538	191	1.026e-06	1.382e-03	SF3B3:15 GEMIN4:101 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292
Peptide Biosynthetic Process (GO:0043043)	-0.11685336	134	3.152e-06	3.397e-03	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 MRPL10:312 RPLP0:328
Macromolecule Biosynthetic Process (GO:0	-0.10402129	159	6.388e-06	5.692e-03	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 LARGE1:276 MRPL10:312
Positive Regulation Of Transcription By	-0.04467808	879	9.623e-06	7.407e-03	NLRP3:2 TXK:3 BCL2L12:32 TRIAP1:33 HAX1:39 HES1:41
RNA Processing (GO:0006396)	-0.09617770	166	2.008e-05	1.082e-02	SF3B3:15 DEDD2:35 NCBP2:106 LSM1:113 EXOSC10:214 NUFP1:230
Inorganic Cation Transmembrane Transport	0.07494351	277	1.930e-05	1.082e-02	TRPM5:2 KCNK15:45 KCNK18:52 TRPM6:103 KCNV2:120 SLC6A6:125
Intermediate Filament Organization (GO:0	0.17474263	50	1.940e-05	1.082e-02	KRT80:228 KRT73:304 KRT32:326 BFSPP2:616 KRT13:1035 PKP1:1076
Positive Regulation Of Cell Motility And Tra	-0.03758071	1160	2.393e-05	1.172e-02	NLRP3:2 TXK:3 SF3B3:15 BCL2L12:32 TRIAP1:33 HAX1:39
Monatomic Cation Transmembrane Transport	0.07354281	273	3.137e-05	1.408e-02	TRPM5:2 KCNK15:45 KCNK18:52 TRPM6:103 KCNV2:120 SLC6A6:125
Vesicle Budding From Membrane (GO:000690	-0.21775513	30	3.677e-05	1.524e-02	SNX3:168 TMED10:183 CHMP7:327 MYO18A:353 S100A10:489 FNBP1:1201
RNA Splicing (GO:0008380)	-0.12260892	86	8.667e-05	3.336e-02	SF3B3:15 NCBP2:106 LSM1:113 MAGOHB:282 LSM4:360 KHSPR:407
B Cell Homeostasis (GO:0001782)	-0.36612130	9	1.428e-04	5.128e-02	GAPT:45 TNFRSF13B:623 TNFAIP3:812 MEFP2C:1250 PPP2R3C:1530 LYN:1816
Amino Acid Transport (GO:0006865)	0.16687718	43	1.543e-04	5.181e-02	SLC43A1:96 SLC6A6:125 SLC6A20:640 SLC6A15:867 SLC36A2:1318 SLC38A1:1446
Translation (GO:0006412)	-0.07644985	206	1.635e-04	5.181e-02	MRPS18C:71 RPS13:196 MRPL34:202 MRPL12:235 MRPL10:312 RPLP0:328
rRNA Metabolic Process (GO:0016072)	-0.12000399	80	2.104e-04	6.297e-02	DEDD2:35 EXOSC10:214 EBNA1BP2:224 NVL:303 EXOSC6:351 RPL27:496
Metal Ion Transport (GO:0030001)	0.08458162	157	2.641e-04	6.776e-02	CLDN16:21 KCNJ16:41 SLC17A4:50 KCNK18:52 SLC17A3:111 ATP2A1:278
Protein Localization To Centrosome (GO:0	0.25636597	17	2.532e-04	6.776e-02	CEP192:131 GOLGB1:434 KIAA0753:633 STL1:690 C2CD3:765 NUDCD3:1054
Regulation Of Small GTPase Mediated Sign	0.10124245	110	2.502e-04	6.776e-02	FGD5:22 TIAM2:82 ARHGEF19:110 SYDE12:180 FAM13B:371 RASGRF1:600
Regulation Of Transcription By RNA Polym	-0.02704334	1713	3.134e-04	7.675e-02	NLRP3:2 TXK:3 SF3B3:15 BRMS1L:17 BCL2L12:32 TRIAP1:33
Monocarboxylic Acid Transport (GO:001571	0.12884845	63	4.097e-04	9.598e-02	SLC27A4:151 ATP8B1:334 SLC16A5:337 SLC5A12:340 TSP02:531 SLC10A2:768
Potassium Ion Transmembrane Transport (G	0.08794635	132	4.882e-04	1.119e-01	KCNJ16:41 KCNK15:45 KCNK18:52 KCNV2:120 LRRCS5:307 KCNF1:580
piRNA Processing (GO:0034587)	0.23332274	18	6.113e-04	1.318e-01	MOV10L1:135 PIWIL1:349 GATC2:562 ANKRD34B:553 PNDC1:1064 EXD1:1322
Lysosome Biogenesis (GO:0032418)	-0.16105531	37	7.027e-04	1.402e-01	TMEM106B:307 FCER1A:745 KIF5B:1195 PIK3CG:1455 TFEF1490 NDEL1:1850
Ribosome Biogenesis (GO:0042518)	-0.08353516	139	6.922e-04	1.402e-01	EIF2S1:47 RPS13:196 EXOSC10:214 EBNA1BP2:224 RPS19BP1:302 NVL:303
Negative Regulation Of Cell Motility And	-0.08677675	125	8.258e-04	1.589e-01	SF3B3:15 BRMS1L:17 TP53NP1:421 AMH:453 SLIT2:589 OGT:790
DNA Replication Checkpoint Signaling (GO	0.245758074	15	9.852e-04	1.659e-01	CLSPN:198 CDC45:381 ORC1:383 DNA2:1172 CDC6:1822 TIMELESS:2283
Negative Regulation Of Cell Migration (GO	-0.07307347	154	9.572e-04	1.659e-01	SF3B3:15 BRMS1L:17 TP53NP1:421 AMH:453 SLIT2:589 CLASP2:707
Sodium Ion Transport (GO:0006814)	0.10391530	85	9.415e-04	1.659e-01	SLC17A4:50 WKNA1:102 SLC17A3:111 SLC6A6:125 SCN10A:289 SLC5A12:340
Vasodilation (GO:0042311)	0.24583491	15	9.803e-04	1.659e-01	ADRB1:740 SOD1:1026 NOS3:1523 NOS1:2871 AGTR2:3209 SOD2:3219
Regulation Of Cell Proliferation And Prolif	-0.03663781	696	1.136e-03	1.855e-01	MTS1R:1 IFI251:29 HES1:41 PCOD10:48 IL31RA:78 IGFBP6:98
Cytoplasmic Translation (GO:0002181)	-0.10832109	73	1.391e-03	2.204e-01	RPS13:196 RPLP0:328 RPS5:452 RPL27:496 RPL6:593 RPL35A:714
Protein Peptidyl-Prolyl Isomerization (G	-0.24543711	14	1.476e-03	2.272e-01	PP1F:863 PP1G:1167 PP1L3:1240 PPLD:1959 PP1C:2243 PIN1:2444
Regulation Of mRNA Splicing, Via Spliceo	-0.10073673	83	1.532e-03	2.293e-01	CELF6:30 SRRM4:375 HNRNP1:475 HNRNPAL1:579 ZBTB7A:697 PTBP1:730
Monocarboxylic Acid Catabolic Process (G	0.20331876	20	1.649e-03	2.339e-01	LPIN1:226 CYP26C1:268 ACOT6:609 HOGA1:712 LPIN2:875 AGXT:1969
Positive Regulation Of Interferon-Beta P	-0.16080771	32	1.649e-03	2.339e-01	IRF3:228 TLR4:416 RIKT3:704 PTPN11:763 IRF5:864 HSP90AA1:962
Regulation Of DNA-templated Transcriptio	-0.02391865	1646	1.723e-03	2.381e-01	SF3B3:15 DEDD2:35 HES1:41 CDK14:44 IRF2:50 YEATS4:61
Positive Regulation Of Transcription Ini	-0.12754104	50	1.822e-03	2.455e-01	NKX2-5:221 TAF10:429 MED7:1021 MED11:1071 MED16:1137 MED29:1148

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
HOUNKPE_HOUSEKEEPING_GENES	-0.073436046	1061	1.198e-15	7.773e-12	STK38:8 CCT2:23 DERL1:24 CMPK1:34 HAX1:39 EIF2S1:47
REACTOME_METABOLISM_OF_RNA	-0.08325670	641	8.579e-13	2.783e-09	SF3B3:15 RAN:16 PSMD8:73 APOBEC1:85 GEMIN4:101 NCBP2:106
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.06132198	1150	4.049e-12	8.756e-09	RSU1:6 LAPTM4B:9 CCT2:23 MAPK9:37 CDK14:44 EIF2S1:47
REACTOME_CELLULAR_RESPONSES_TO_STIMULI	-0.07732462	692	5.598e-12	9.079e-09	NLRP3:2 CCS:27 DEDD2:35 MAPK9:37 EIF2S1:47 LMNB1:52
REACTOME_REGULATION_OF_MRNA_STABILITY_BY	-0.20730775	83	6.798e-11	8.821e-08	PSMD8:73 ELAVL1:142 PSMB5:223 ANP32A:261 HSPA8:292 PSMA1:344
REACTOME_THE_ROLE_OF_GTSE1_IN_G2_M_PROGR	-0.21648249	73	1.635e-10	1.769e-07	PSMD8:73 MAPRE1:134 PSMB5:223 TUBB2B:299 PSMA1:344 PSMC3:451
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	-0.05976798	1011	1.913e-10	1.773e-07	TRIAP1:33 HES1:41 YEATS4:61 GANT:64 CDC27:68 NBN:69
REACTOME_INFECTIONOUS_DISEASE	-0.06601246	809	2.378e-10	1.929e-07	NLRP3:2 RAN:16 TAB2:19 PSMD8:73 GEMIN4:101 NCBP2:106
REACTOME_REGULATION_OF_EXPRESSION_OF_SL	-0.14962606	146	4.559e-10	3.286e-07	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
KIM_BIPOLAR_DISORDER_OLGODENDROCYTE_DISE	-0.1076215	624	1.967e-09	1.276e-06	NSF:7 RAN:16 CLTB:18 CCT2:23 MAPK9:37 PPP6C:55
REACTOME_SIGNALING_BY_ROBO_RECEPTORS	-0.12426449	189	4.051e-09	2.389e-06	PSMD8:73 NCBP2:106 RPS13:196 PSMB5:223 ELOB:231 MAGOHB:282
DIAZ_CHRONIC_MYELOGENOUS_LEUKEMIA_UP	-0.04746257	1319	1.094e-08	5.913e-06	RSU1:6 LAPTM4B:9 SF3B3:15 CCT2:23 CMPK1:34 KCTD3:42
REACTOME_SIGNALING_BY_THE_B_CELL_RECEPTO	-0.15819368	105	2.189e-08	1.063e-05	PSMD8:73 BLK:147 ORAI1:217 PSMB5:223 PSMA1:344 SH3KBP1:422
PUJANA_BRCA1_PCC_NETWORK	-0.04406881	1479	2.294e-08	1.063e-05	ITPK1:11 RAN:16 CCT2:23 TRIAP1:33 CHST7:43 EIF2S1:47
MILI_PSEUDOPODIA_HAPTOTOXIS_UP	-0.07403883	485	2.690e-08	1.141e-05	PCDD10:48 MRPS18C:71 STYX:128 PIAR1:143 TIMM8A:195 SRP54:197
REACTOME_CLECTA_DECTIN_1_SIGNALING	-0.16582669	94	2.815e-08	1.141e-05	TAB2:19 PSMD8:73 PSMB5:223 CLEC7A:320 PSMA1:344 PSMC3:451
REACTOME_UCH7_PROTEINASES	-0.17278374	85	3.730e-08	1.407e-05	PSMD8:73 TPST1:129 PSMB5:223 PSMA1:344 PSMC3:451 USP15:517
BORCZUK_MALIGNANT_MESOTHELIOMA_UP	-0.09434338	288	3.905e-08	1.407e-05	RAN:16 CCT2:23 TRIAP1:33 MAPK9:37 CDC27:68 PSMD8:73
ACEVEDO_METHYLATED_IN_LIVER_CANCER_DN	0.06362738	637	5.063e-08	1.729e-05	FGD5:22 ADGRV1:31 ADGRG7:28 ANK1:42 TIAM2:82 SLC43A1:96
REACTOME_MRNA_SPLICING	-0.11347883	193	5.718e-08	1.855e-05	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_HIV_INFECTION	-0.10597977	214	9.619e-08	2.837e-05	RAN:16 PSMD8:73 NCBP2:106 PSMB5:223 ELOB:231 CDDB8:239
HSIAO_HOUSEKEEPING_GENES	-0.08545768	332	9.467e-08	2.837e-05	ITPK1:11 PSMD8:73 TMED10:183 PSMB5:223 COX41:287 RPLP0:328
REACTOME_FCR1I_MEDIATED_NF_KB_ACTIVATION	-0.17521528	77	1.079e-07	3.044e-05	TAB2:19 PSMD8:73 PSMB5:223 FCER1A:724 PSMA1:344 PSMC3:451
REACTOME_RUNX1_REGULATES_TRANSCRIPTION_O	-0.16348035	88	1.174e-07	3.175e-05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 H4C1:393 PSMC3:451
PUJANA_CHEK2_PCC_NETWORK	-0.05850112	704	1.478e-07	3.836e-05	RAN:16 CCT2:23 TRIAP1:33 EIF2S1:47 PCDD10:48 LMNB1:52
REACTOME_SCF_BETA_TRCP_MEDIATED_DEGRADAT	-0.20898914	52	1.655e-07	4.003e-05	PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DNA_REPLICATION_PRE_INITIATION	-0.14200887	114	1.666e-07	4.003e-05	CDC27:68 PSMD8:73 PSMB5:223 PSMA1:344 H4C1:393 PSMC3:451
REACTOME_DOWNSTREAM_SIGNALING_EVENTS_OF_	-0.17345803	76	1.737e-07	4.025e-05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 NFKBIA:799
WANG_TUMOR_INVASIVENESS_UP	-0.08111609	351	1.937e-07	4.201e-05	CCT2:23 STX7:72 NHP2:121 ELAVL1:142 SCAF8:145 COX41:287
REACTOME_C_TYPE_LECTIN_RECEPTORS_CLRS	-0.13942790	117	1.943e-07	4.201e-05	TAB2:19 PSMD8:73 PSMB5:223 FCER1G:274 CLEC7A:320 PSMA1:344
REACTOME_INNATE_IMMUNE_SYSTEM	-0.05235453	865	2.157e-07	4.488e-05	NLRP3:2 TXK:3 TAB2:19 CCT2:23 MAPK9:37 VPS35L:40
REACTOME_APC_C_MEDIATED_DEGRADATION_OF_C	-0.16458800	83	2.214e-07	4.488e-05	CDC27:68 PSMD8:73 CDC20:111 PSMB5:223 PSMA1:344 PSMC3:451
REACTOME_DECTIN_1_MEDIATED_NONCANONICAL_	-0.19534626	58	2.691e-07	5.146e-05	PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566 PSMA8:801
KEGG_SPLICEOSOME	-0.14271748	109	2.697e-07	5.146e-05	SF3B3:15 NCBP2:106 SNRPB:249 MAGOHB:282 HSPA8:292 LSM4:360
REACTOME_TCR_SIGNALING	-0.14409078	106	3.031e-07	5.314e-05	TAB2:19 PSMD8:73 PSMB5:223 PSMA1:344 PSMC3:451 PSMA4:566
REACTOME_DISEASES_OF_CELLULAR_TRANSDUCTION	-0.07359314	415	2.986e-07	5.314e-05	HES1:41 PSMD8:73 NCBP2:106 CNKSR2:118 KREMLIN1:131 NR4A1:174
BENPORATH_MYC_MAX_TARGETS	-0.05756673	691	2.966e-07	5.314e-05	MTS1R:1 HAX1:39 EIF2S1:47 PCDD10:48 PPP6C:55 STX7:72
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_R	-0.10892158	186	3.133e-07	5.349e-05	PSMD8:73 BLK:147 PSMB5:223 PSMA1:344 CDK7:361 H4C1:393
REACTOME_MITOTIC_G1_PHASE_AND_G1_S_TRANS	-0.12342766	144	3.271e-07	5.442e-05	PSMD8:73 PSMB5:223 PSMA1:344 CDK7:361 PSMC3:451 MYBL2:553
MARTENS_RETINOIN_RESPONSE_DN	-0.05691750	699	3.475e-07	5.636e-05	RAN:16 TRIAP1:33 HES1:41 MRPS18C:71 PEX12:89 GEMIN4:101

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
22q11 Deletion Syndrome	0.28414095	19	1.810e-05	5.919e-02	HIRA:85 MJMD1C:603 CHR6:671 ZDHHC8:1370 DGCR8:1503 COMT:1536
Glioma	-0.03194195	1776	1.495e-05	5.919e-02	MST1R:1 RSU1:6 BCL2L12:32 CMPK1:34 MAPK9:37 HES1:41
Tetany	0.28378849	21	6.759e-06	5.919e-02	CLDN16:21 HIRA:85 TRPM6:103 MJMD1C:603 KCNJ1:644 SLC12A3:767
Glioblastoma	-0.03110326	1593	5.686e-05	1.116e-01	RSU1:6 BCL2L12:32 MAPK9:37 HES1:41 PCOD10:48 RARRES2:57
Precursor T-Cell Lymphoblastic Leukemia-Neoplasm Metastasis	-0.06466747	337	4.494e-05	1.116e-01	HES1:41 ZEB2:86 IGFBP7:119 TIMMBA:195 NKX2-5:221 CSA2:226
Anemia, hereditary spherocytic hemolytic	0.48861271	5	1.544e-04	1.515e-01	MST1R:1 RSU1:6 LAPTM4B:9 RAN:16 BRMS1L:17 COL6A3:21
Carcinogenesis	-0.02209973	3287	1.527e-04	1.515e-01	ANK1:42 SPTA1:70 SLC4A11:30 EPB42:275 SPTB:394 NA
Classical Hodgkin's Lymphoma	-0.09339394	142	1.263e-04	1.515e-01	MST1R:1 FSCN1:210 CYS1LFR1:238 TNFRSF17:315 EPB42:454 IL3RA:567
Spherocytosis	0.48861271	5	1.544e-04	1.515e-01	ANK1:42 SPTA1:70 SLC4A11:30 EPB42:275 SPTB:394 NA
Cytochrome-c Oxidase Deficiency	-0.17963093	36	1.929e-04	1.582e-01	CCS:27 COAT:136 COA6:159 COX41:287 ETHE1:592 COX5A:973
Lymphoma	-0.03432349	1072	1.936e-04	1.582e-01	MST1R:1 NLRP3:2 NSF:7 STK38:8 NBN:69 TNFSF13B:139
Malignant neoplasm of stomach	-0.02652887	1876	2.342e-04	1.767e-01	MST1R:1 STK38:8 LAPTM4B:9 COL6A3:21 CDK14:44 EIF2S1:47
Central neuroblastoma	-0.02918871	1407	3.443e-04	2.252e-01	NSF:7 RAN:16 MAPK9:37 HES1:41 IRF2:50 IL31RA:78
Squamous cell carcinoma	-0.02843021	1500	3.369e-04	2.252e-01	MST1R:1 MAPK9:37 HAX1:39 HES1:41 CDK14:44 EIF2S1:47
HIV Infections	-0.04054014	634	5.652e-04	3.465e-01	RAN:16 CMPK1:34 MAPK9:37 NBN:69 PSMD8:73 CCL19:179
22q11 partial monosomy syndrome	0.29166078	11	8.100e-04	3.484e-01	HIRA:85 MJMD1C:603 COMT:1536 ARVCF:1658 PI4KA:2297 TBL1:2862
Bilateral cataracts (disorder)	0.09504775	103	8.745e-04	3.484e-01	GSTO1:117 CRYGA:638 SMN2:499 BFSPP2:616 NFE2L2:664 SMN1:676
Broad chest	-0.35930702	7	9.945e-04	3.484e-01	BRAF:526 MAP2K1:675 PTPN11:763 CHST3:2186 DYM:3635 TRAPPCC2:3870
Familial intrahepatic cholestasis of pre	0.39443020	6	8.200e-04	3.484e-01	ATP8B1:334 ABCB8A:1161 NR1H4:1321 VCAM1:1624 ABCB11:1746 TNF:3765
Hamantoma of tongue	0.42522236	5	9.885e-04	3.484e-01	OFD1:692 C2CD3:765 WDPCP:1022 TCTN3:1434 NEK1:1923
Leukemogenesis	-0.09994559	617	7.974e-04	3.484e-01	HAX1:39 HES1:41 NR4A1:174 NKX2-5:221 CDDB8:239 PRKAA2:255
Metabolic Bone Disorder	0.17754689	30	7.664e-04	3.484e-01	CA2:459 CYP2R1:482 PTH:1027 ESR1:1070 IL6:1085 MMP2:1182
Neuroblastoma	-0.02659929	1445	9.663e-04	3.484e-01	NSF:7 RAN:16 CMPK1:34 MAPK9:37 HES1